NRES 746, Week 10, Lecture 2: Ridge Is a Prior

Answer key. Answers are in magenta sans-serif type; longer answers have a bar in the margin.

Instructions: Work alone on Part 1 (about 4 minutes), then compare with a neighbor for Part 2. Keep laptops closed.

The setup

Bird counts at n survey routes give each route's change in log abundance since last year, $y_1, \ldots, y_n$. Model them as $y_i \sim \text{Normal}(\mu, \sigma)$ with σ known; μ is the regional trend.

- **Monday's ridge fit** minimizes the penalized NLL

$$\text{PNLL}(\mu) = \sum_{i=1}^{n} \frac{(y_i - \mu)^2}{2\sigma^2} + \lambda\mu^2$$

- **A Bayesian fit** gives μ a Normal(0, τ) prior. Up to a constant, the negative log of that prior is $\mu^2/(2\tau^2)$.

Part 1: Match them (individual)

1a. Write the negative log posterior for μ, dropping constants (recall: posterior $\propto$ likelihood $\times$ prior). For what value of λ is it the same function as PNLL(μ)?

$$-\log P(\mu \mid y) = \sum_{i=1}^{n} \frac{(y_i - \mu)^2}{2\sigma^2} + \frac{\mu^2}{2\tau^2} + \text{const.}$$

$\lambda = 1/(2\tau^2)$

1b. So the ridge estimate is which point of the posterior? What kind of prior (large or small τ) gives ordinary maximum likelihood?

The posterior mode (the peak of the posterior). A very vague prior (large τ, so $\lambda \to 0$) gives the MLE.

Part 2: The lasso (with a neighbor)

2a. Monday's lasso penalty is $\lambda|\mu|$. What prior density for μ does it correspond to? Sketch its shape on the axes, with a Normal prior for comparison.

Prior: Laplace (double exponential), density $\propto e^{-\lambda|\mu|}$: a sharp peak at 0 and heavier tails than a Normal.

